
CofactVLA: Deconfounding Vision-Language-Action Models via Counterfactual Intervention

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Abstract

Vision-Language-Action (VLA) models have driven significant progress in robotic manipulation, yet they fundamentally struggle with the “vision-override” phenomenon. Driven by the severe modality imbalance between dense visual streams and sparse linguistic instructions, VLAs frequently fall prey to causal confusion. Instead of treating language as the primary causal driver, the policy entirely bypasses the original instruction by overfitting to spurious visual confounders, such as prominent objects or familiar layouts. To systematically alleviate this bias, we formalize the process of action generation as a Dual-path Deconfounding Graph (DDG) and propose **CofactVLA**, a novel causal intervention framework. By dynamically constructing a language-masked counterfactual branch within a single forward pass, CofactVLA isolates and neutralizes visual confounders through two synergistic mechanisms. First, Action-Level Orthogonal Projection Guidance (OPG) geometrically projects the factual velocity field away from the counterfactual visual bias during continuous flow matching, extracting the pure semantic intent. Second, Feature-Level Counterfactual Covariance Reduction (CCR) mathematically deconfounds latent representations by penalizing the positive eigenspace of the covariance difference, explicitly suppressing dominant visual shortcuts while preserving the causal language intent. Extensive experiments demonstrate that CofactVLA establishes a new state-of-the-art across diverse simulation benchmarks. Beyond simulation, real-world robot experiments demonstrate the causal efficacy of our method in bridging the generalization gap, yielding a 52.3% absolute success rate gain under out-of-distribution scenarios.

1 Introduction

The integration of pre-trained Vision-Language Models (VLMs) into robotic control has driven remarkable progress in Vision-Language-Action (VLA) models[1, 2, 3, 4, 5, 6, 7, 8, 9]. However, the deployment of VLAs in open-world environments reveals a critical vulnerability known as the **Vision-Override Phenomenon**. Due to the severe modality imbalance in robotics datasets—where dense visual streams inevitably dominate sparse linguistic instructions—VLA models frequently suffer from **causal confusion**, as shown in Figure 1(a). Instead of treating the language instruction as

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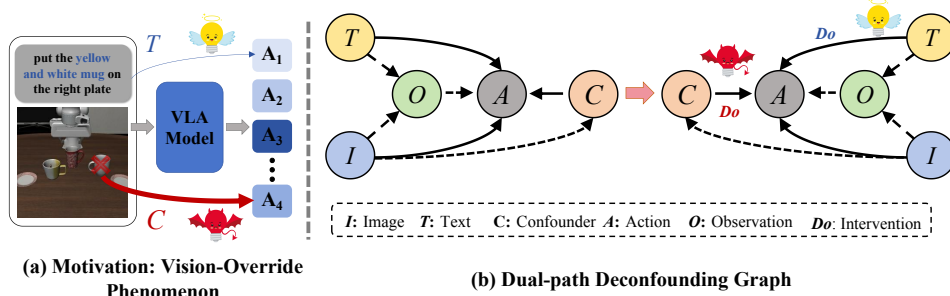


Figure 1: **(a) Motivation:** Visual modality inadvertently induces a spurious backdoor path ($I \dashrightarrow C \rightarrow A$). The visual confounder C acts as a shortcut that dictates the action A , overriding the true semantic intent T . **(b) Our proposed Dual-path Deconfounding Graph:** The red *do*-operator signifies Feature-Level Counterfactual Covariance Reduction (CCR), which explicitly suppresses the spurious visual confounder C . The blue *do*-operator signifies Action-Level Orthogonal Projection Guidance (OPG), which geometrically extracts and enhances the pure language intent T .

the causal driver, the model overfits to spurious visual confounders, such as habitually grasping the most salient object while entirely ignoring the text prompt.

Recent efforts to address the vision-override phenomenon generally fall into two categories. Data-centric approaches (e.g., linguistic rephrasing [10] or data augmentation in specific domains [11, 12]) are notoriously difficult to scale across open-world tasks, thereby failing to systematically decouple entangled representations. At the action level, inference strategies like Classifier-Free Guidance [13] attempt to suppress visual priors via dual-branch scalar subtraction. However, this post-hoc linear extrapolation often inflates unaligned noise, pushing continuous robotic trajectories out-of-distribution and causing unsafe executions. More importantly, these methods lack a principled mechanism to address structural modality collapse within the model’s internal representations.

To this end, we approach the vision-override problem through the lens of **Counterfactual Intervention**. We first formalize the VLA decision-making process as a **Dual-path Deconfounding Graph (DDG)**, as shown in Figure 1(b). Ideally, the robotic action A should be properly conditioned on the language instruction T and the image observation I . However, due to the severe modality imbalance in training data, the model inadvertently learns a spurious causal path $I \dashrightarrow C \rightarrow A$. Here, C represents the visual confounder (e.g., prominent out-of-context objects or familiar scene layouts). This confounder acts as a dominant visual shortcut that heavily dictates the action A while bypassing the text instruction T . To eliminate the spurious effect of C and recover the true causal effect of the language instruction, we formulate a counterfactual question: “*What action would the VLA model inherently take in this exact visual scene if the language instruction were entirely absent?*” By explicitly constructing a language-masked counterfactual branch within a single forward pass, we dynamically isolate the pure visual confounder C . This allows us to perform end-to-end causal interventions to sever the spurious path (the red *do*-operator on $C \rightarrow A$) and protect the pure language intent (the blue *do*-operator on $T \rightarrow A$).

To summarize, our main contributions are as follows:

- **A Unified Deconfounding Framework for VLAs:** We propose CofactVLA, which formulates the vision-override phenomenon as a causal confusion problem. By constructing the action generation process as a DDG, we introduce a language-masked counterfactual branch within a single forward pass, which aims to isolate spurious visual confounders.
- **Action-Level Orthogonal Projection Guidance (OPG):** We introduce an action-level geometric decoupling strategy for continuous flow matching, which projects the factual velocity field orthogonally away from the counterfactual visual bias to extract the pure semantic intent, yielding strictly deconfounded action trajectories.
- **Feature-Level Counterfactual Covariance Reduction (CCR):** To explicitly mitigate the interference of visual confounders at the feature level, we introduce CCR by penalizing the positive eigenspace of the covariance difference between the counterfactual and factual.
- **State-of-the-Art Performance and OOD Generalization:** CofactVLA achieves highly competitive, often state-of-the-art results across simulated and real-world physical deploy-

ments, while exhibiting robust zero-shot generalization in unseen Out-of-Distribution (OOD) scenarios.

2 Related Work

2.1 Vision-Language-Action Models

Vision-Language-Action (VLA) models have emerged as powerful end-to-end visuomotor policies that map raw visual observations and natural-language instructions directly to low-level control actions. Typically, these models adapt large-scale pre-trained Vision-Language Models (VLMs) via Supervised Fine-Tuning (SFT) or Reinforcement Learning (RL) [1, 2, 14, 15]. A predominant paradigm formulates continuous control as a sequence modeling task, discretizing action chunks and employing autoregressive token decoding to inherit the scalability and open-vocabulary generalization of foundation models [1, 2, 14, 16, 17]. To enhance deployment practicality and generalization, subsequent researches have explored optimized fine-tuning strategies [18], parameter-efficient adaptation at smaller scales [19, 20, 21], improved perceptual grounding [22], and continual post-training from experience [23]. Concurrently, to alleviate the latency bottleneck of autoregressive generation, efficiency-oriented VLAs have investigated parallel decoding and system-level accelerations, such as token compression, speculative execution, and extreme quantization [18, 24, 25, 26].

Beyond autoregressive paradigms, diffusion and flow-matching frameworks have gained traction for modeling complex, multi-modal continuous action distributions, synthesizing action chunks via iterative denoising or ordinary differential equation (ODE) integration [27, 28, 29, 30, 31]. Recent advancements have successfully scaled these continuous formulations, utilizing distillation to accelerate sampling [32, 33], while flow-matching-based VLAs and hybrid diffusion-autoregressive architectures explicitly target high-frequency, generalist robotic control [34, 35, 17, 16].

2.2 Causality and Deconfounding in Robotic Manipulation

Causal inference has increasingly garnered attention for mitigating distribution shifts and enhancing the robustness [36, 37]. While early robotics efforts injected causal priors into imitation frameworks [38, 39, 28], recent language-conditioned VLAs primarily rely on counterfactual data augmentation to reduce spurious visual shortcuts [11, 14]. However, these data-centric methods inherently struggle to scale across open-world scenarios and fail to fundamentally disentangle collapsed internal representations. In contrast, our proposed CofactVLA explicitly formulates the action generation process as a **Dual-path Deconfounding Graph**. Guided by this structure, we execute a systematic dual-path intervention: operating simultaneously at the latent feature level to suppress visual confounders, and at the continuous action level to geometrically extract the pure semantic intent.

3 Methods

3.1 A Causal View of VLA

VLA models aim to learn a continuous policy $\pi(A \mid O, T)$ that maps multimodal observations O and language instructions T to actions A via generative processes (e.g., flow matching). We formalize this decision-making process through structural causal model, as illustrated in Figure 1. Ideally, A should be causally driven by the semantic intent T and grounded by O . However, due to the severe modality imbalance in training data, the model inadvertently learns a spurious backdoor path $I \dashrightarrow C \rightarrow A$. Here, C represents the latent visual confounder (e.g., prominent out-of-context objects or familiar layouts) that heavily dictates the action A and overrides the causal effect of T .

To mitigate the spurious effect of C and recover the true causal effect of T , we pose a counterfactual question: “*What action would the VLA inherently take if the language instruction were entirely absent?*” To operationalize this, we propose a language-masked dual-branch framework by combining a factual branch (conditioned on O and T) with a counterfactual branch (conditioned on O). Thus, its generated representations and actions are driven exclusively by its inherent visual biases for the given scene. Building upon this design, CofactVLA deconfounds the model through two complementary interventions (the *do*-operators in Fig. 2), detailed in the following sections.

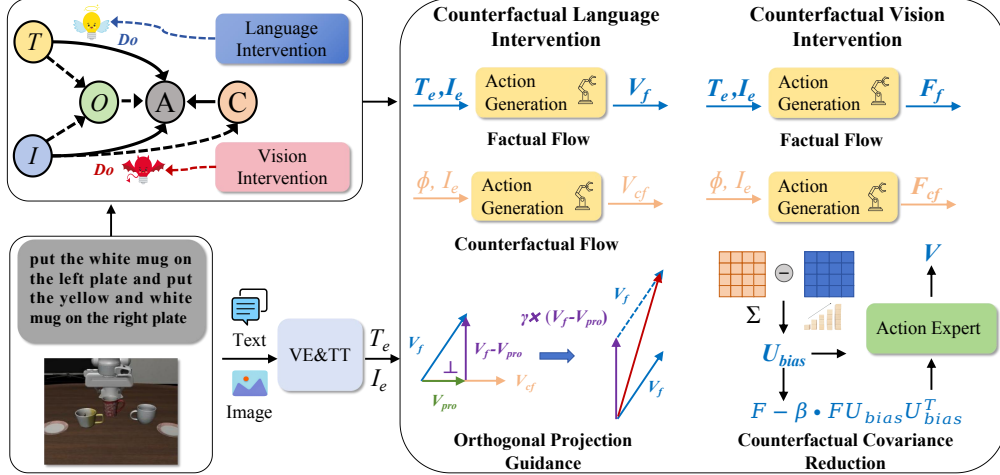


Figure 2: **The overall architecture of CofactVLA.** (Left) Driven by the proposed Dual-path Deconfounding Graph (DDG), we propose an intervention framework to address the vision-override phenomenon. (Right) The framework consists of two core components dynamically built upon factual and counterfactual flows. To execute Language Intervention, Orthogonal Projection Guidance (OPG) geometrically extracts the pure semantic intent by projecting the factual flow V_f onto the counterfactual visual prior V_{cf} . To execute Vision Intervention, Contrastive Covariance Reduction (CCR) extracts the dominant visual confounder bias U_{bias} via the covariance difference of latent features to cleanse the features before feeding them to the action expert.

3.2 Action-Level Deconfounding: Orthogonal Projection Guidance (OPG)

Building upon the established counterfactual anchor, we first introduce our causal intervention at the action level. A fundamental challenge in robotic control lies in its inherently generative and multi-valued: a single semantic intent T could be successfully executed via multiple valid trajectories.

From a probabilistic perspective, let $\mathcal{A}^H$ be the continuous action-chunk space with action horizon H , partitioned into discrete valid modes $Z = h(A)$ (e.g., different grasping styles or approach angles). Even if the model perfectly perceives the text instruction T , the multi-valued nature of the expert demonstrations allows the visual confounder C to spuriously bias the mode selection. Formally, if the demonstrations exhibit nuisance-correlated mode selection $I(Z; C | O) > 0$, data processing inequalities yield $I(A; C | O) > 0$, where $I(\cdot; \cdot | \cdot)$ represents the conditional mutual information. This induces a strict negative log-likelihood (NLL) gap under distribution shifts:

$$H(A | O) - H(A | O, C) = I(A; C | O) > 0, \quad (1)$$

where $H(\cdot | \cdot)$ denotes conditional entropy. This theorem explicitly dictates that the visual confounder intrinsically corrupts the multi-modal distribution of the expert policy. To eradicate this bias without destroying the valid action manifold, an explicit intervention is required at the generative output space to re-weight the mode selection probabilities.

We formulate this intervention within the continuous flow matching framework. Let $p_{cond}^\tau := p(A^\tau | O)$ and $p_{uncond}^\tau := p(A^\tau | T)$ denote the marginal densities of the factual and counterfactual branches at diffusion time τ , respectively. To extract the pure semantic intent, we aim to perform a Product-of-Experts (PoE) composition in the probability density space. Crucially, under the Gaussian forward process $A^\tau = \tau A + (1 - \tau)\omega$, the score function $\nabla_{A^\tau} \log p^\tau$ is linearly convertible from the predicted flow-matching velocity field $v_\theta(A^\tau, \tau | \cdot)$:

$$\nabla_{A^\tau} \log p^\tau(A^\tau | \cdot) = -\frac{A^\tau + \tau v_\theta(A^\tau, \tau | \cdot)}{1 - \tau}. \quad (2)$$

Because of this exact linear equivalence, manipulating the generated velocity fields mathematically corresponds to composing the causal score functions. Counterfactual action guidance [40] attempts this via simple scalar extrapolation: $v_{CFG} = v_{uncond} + \gamma(v_{cond} - v_{uncond})$. However, in the high-dimensional non-linear action space, this blind scalar subtraction assumes orthogonal separability. When this assumption fails, it exponentially inflates unaligned visual noise, pushing the generated

flow out-of-distribution and causing unsafe physical executions. To robustly perform the causal composition, we propose OPG. Let v_{cond} and v_{uncond} be the predicted velocity fields from the factual and counterfactual branches. We geometrically project the factual field onto the counterfactual visual instinct:

$$v_{proj} = \frac{\langle v_{cond}, v_{uncond} \rangle}{\|v_{uncond}\|_2^2 + \epsilon} v_{uncond}. \quad (3)$$

By explicitly subtracting this collinear visual bias, we extract the strictly orthogonal velocity component $v_{\perp}$, which represents the pure, deconfounded language semantic intent:

$$v_{\perp} = v_{cond} - v_{proj}. \quad (4)$$

The final causally intervened velocity field is then constructed as:

$$v_{causal} = v_{cond} + \gamma \cdot v_{\perp}, \quad (5)$$

where $\gamma > 1.0$ is the guidance scale. By virtue of the score equivalence, this orthogonalization guarantees that our intervention exclusively re-weights the causal mode odds along the semantic direction, cleanly eradicating visual bias while ensuring the generated trajectories adhere strictly to the valid robotic action manifold.

3.3 Feature-Level Deconfounding: Counterfactual Covariance Reduction (CCR)

While OPG robustly corrects the multi-modal action distribution at the output level, the root cause of the vision-override phenomenon originates within the latent representation space. Due to the severe modality imbalance during training, visual confounders within the VLA backbone often dominate the attention mechanisms, establishing spurious shortcuts in the Key-Value (KV) cache before the action generator even receives the features. To systematically eradicate this, we propose Counterfactual Covariance Reduction (CCR), a causal intervention applied directly to the latent attention features.

Let F_f and $F_{cf} \in \mathbb{R}^{N \times d}$ denote the flattened valid attention features (e.g., Keys and Values) from the transformer layers of the factual and counterfactual branches, respectively. We compute their centered empirical covariance matrices: $\Sigma_f = \text{Cov}(F_f)$ and $\Sigma_{cf} = \text{Cov}(F_{cf})$, and formulate our intervention based on the covariance difference:

$$\Delta\Sigma = \Sigma_{cf} - \Sigma_f \quad (6)$$

To mathematically justify how this matrix isolates the visual bias, we introduce a gain-bias decomposition and a contrastive eigengap that separates nuisance- and observation-related directions.

Assumption 1 (Gain-bias decomposition on the interface) *The latent feature space $\mathbb{R}^{d_r}$ could be orthogonally decomposed into two structural subspaces: the causal observation-driven intent subspace $\mathcal{S}_O \subset \mathbb{R}^{d_r}$ and the spurious visual confounder subspace $\mathcal{S}_C \subset \mathbb{R}^{d_r}$, i.e., $\mathbb{R}^{d_r} = \mathcal{S}_O \oplus \mathcal{S}_C$. There exist measurable functions h_O, h_C almost surely*

$$\tilde{\Delta} = \tilde{\Delta}_O + \tilde{\Delta}_C, \tilde{\Delta}_O = h_O(O) \in \mathcal{S}_O, \tilde{\Delta}_C = h_C(C) \in \mathcal{S}_C. \quad (7)$$

Assumption 2 (Contrastive eigengap) *Let $M := \Sigma_0^{-1/2} \Sigma_{\Delta} \Sigma_0^{-1/2}$. The eigenvalues of M restricted to $\Sigma_0^{1/2} \mathcal{S}_C$ are strictly larger than those restricted to $\Sigma_0^{1/2} \mathcal{S}_O$:*

$$\lambda_{\min}(M|_{\Sigma_0^{1/2} \mathcal{S}_C}) > \lambda_{\max}(M|_{\Sigma_0^{1/2} \mathcal{S}_O}). \quad (8)$$

Then, the visual confounder is geometrically identifiable. Let $\Delta\Sigma = U\Lambda U^{\top}$ be the eigendecomposition of the covariance difference. We isolate the positive eigenspace by selecting the top- k eigenvectors corresponding to $\lambda > \epsilon > 0$, constructing the nuisance basis $U_{bias} \in \mathbb{R}^{d \times k}$. First, we project the latent features orthogonally away from U_{bias} to eliminate the visual confounder while strictly preserving the causal language semantics located in the orthogonal negative eigenspace.

Theorem 1 (Delete nuisance bias, keep content gain) *By extracting the principal components of the positive spectrum of $\Delta\Sigma$, U_{bias} spans exactly $\mathcal{S}_C$. The causal feature F_{causal} is obtained by subtracting the collinear nuisance projection from the original features:*

$$F_{causal} = F + \Delta_O = F - \beta(FU_{bias})U_{bias}^{\top}, \quad (9)$$

where $\beta \in (0, 1]$ controls the intervention strength.

Please refer to the Appendix A.1 for methodological proofs. Since U_{bias} is orthogonal to $\mathcal{S}_O$, the semantic language representations remain unperturbed ($F_{causal} \perp \mathcal{S}_C$). In practice, we intercept the computation of the Key and Value states in the final layers of the VLM.

4 Experiments

4.1 Experimental Setup

We evaluate CofactVLA in both **simulation** and **real-world robot environments**. For simulation, we build our model upon the $\pi_{0.5}$ architecture [17] and evaluate it on the standard LIBERO suite [41], as well as the LIBERO-Plus [10] benchmarks to systematically assess zero-shot generalization and robustness against environmental perturbations. For physical deployments, we employ a 6-DoF AgileX PiPer robotic arm equipped with overhead and wrist Intel RealSense cameras operating at 10 Hz. To strictly test the policy’s causal grounding against visual distractors, we design four distinct tasks: *I. remove the cuboid from the blue plate*, *II. move the tennis ball from the yellow plate to the blue plate*, *III. put the apple on the yellow plate*, and *IV. pick the red cube into the yellow plate*. For each task, we employ about 100 expert trajectories for training and 100 independent trials for evaluation. Comprehensive details regarding benchmark metrics, training configurations, and hardware setups are provided in Appendixes A.2 and A.3.

4.2 Main Results on Simulation Benchmarks

As shown in Tables 1-2, we present the quantitative evaluation of CofactVLA against state-of-the-art baselines on the standard LIBERO, the highly challenging LIBERO-Plus benchmarks. Note that the best and second-best results are highlighted in **bold** and underlined, and the methods in Table 2 are trained exclusively on the standard LIBERO dataset to evaluate zero-shot generalization.

Performance on Standard LIBERO.

As shown in Table 1, CofactVLA achieves a state-of-the-art average success rate of **98.5%**, reaching **100.0%** on the Object suite and **99.0%** on the Spatial suite. This confirms that our dual-path intervention effectively enhances fundamental manipulation capabilities, yielding higher execution precision than the base model $\pi_{0.5}$ (96.9%) and the SOTA models X-VLA (98.1%).

Table 1: **Performance comparison (%)** of different VLA models on the **LIBERO benchmark**.

Model	Spatial	Object	Goal	Long	Average
OpenVLA-OFT[18]	97.6	98.4	97.9	94.5	97.1
π_0 [34]	96.8	<u>98.8</u>	95.8	85.2	94.2
$\pi_{0.5}$ [17]	98.8	98.2	98.0	92.4	96.9
DreamVLA[42]	97.5	94.0	89.5	89.5	92.6
X-VLA[43]	<u>98.2</u>	98.6	97.8	97.6	<u>98.1</u>
CofactVLA	99.0	100.0	98.0	<u>97.0</u>	98.5

Table 2: **Zero-shot performance (%)** of different VLA models on the **LIBERO-Plus benchmark**.

Method	Camera	Robot	Language	Light	Background	Noise	Layout	Total
OpenVLA [14]	0.8	3.5	23.0	8.1	34.8	15.2	28.5	15.6
NORA [44]	2.2	37.0	65.1	45.7	58.6	12.8	62.1	39.0
WorldVLA [45]	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
UniVLA [46]	1.8	<u>46.2</u>	69.6	69.0	81.0	21.2	31.9	42.9
π_0 [34]	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
π_0 -Fast [47]	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
OpenVLA-OFT_w [18]	10.4	38.7	70.5	76.8	93.6	49.9	69.9	55.8
Openvla-OFT_m [18]	<u>55.6</u>	21.7	81.0	92.7	<u>91.0</u>	78.6	68.7	67.9
CofactVLA	44.7	49.7	<u>71.8</u>	<u>85.6</u>	83.6	<u>78.0</u>	70.2	69.1

OOD Generalization and Robustness. We evaluate out-of-distribution capabilities on the LIBERO-Plus benchmark (Table 2), which introduces seven distinct visual and environmental perturbations. CofactVLA demonstrates exceptional robustness against these distribution shifts, achieving the highest total success rate of **69.1%** and significantly outperforming the foundational π_0 baseline (53.6%). CofactVLA performs particularly well under novel robotic embodiments (**Robot, 49.7%**) and unseen spatial configurations (**Layout, 70.2%**), scenarios where standard models often overfit to familiar visual cues and fail. Furthermore, it maintains highly competitive performance across lighting, background, and noise variations. Overall, these results indicate that explicitly isolating visual biases practically improves the model’s ability to seamlessly generalize to novel environments and dynamic physical perturbations.

4.3 Main Results on Real-World Robot

As illustrated in Figure 3, we rigorously evaluate the policies in both standard and out-of-distribution (OOD) physical setups. In the standard environments (Figure 3a), CofactVLA achieves a superior average success rate of **90.8%**, outperforming the baseline $\pi_{0.5}$ (**71.0%**). The performance discrepancy across specific tasks strongly highlights the necessity of our deconfounding mechanisms against the “vision-override” phenomenon. For instance, in Task IV—which involves complex visual distractors— $\pi_{0.5}$ struggles significantly, scoring only **31%**. In contrast, CofactVLA maintains precise semantic grounding to achieve **88%**.

The vulnerability of the baseline is further exposed under environmental distribution shifts (Figure 3b). While $\pi_{0.5}$ perfectly solves straightforward tasks (e.g., Task II, **100%**) in familiar settings, it completely collapses (**0%**) when transferred to unseen OOD scenarios. Its overall average plummets to a mere **23.5%**, highlighting a severe reliance on memorized visual shortcuts. Conversely, CofactVLA effectively prevents overfitting to these spurious confounders, demonstrating remarkable zero-shot resilience. It achieves an average success rate of **75.8%** under OOD conditions, marking a massive **+52.3%** absolute improvement over $\pi_{0.5}$ (Please refer to Figures 6-7 and Appendix A.4 for qualitative details). These real-world experiments explicitly confirm that successfully decoupling visual biases inherently bridges the generalization gap, empowering the robot to execute safe, language-grounded manipulation in complex physical environments.

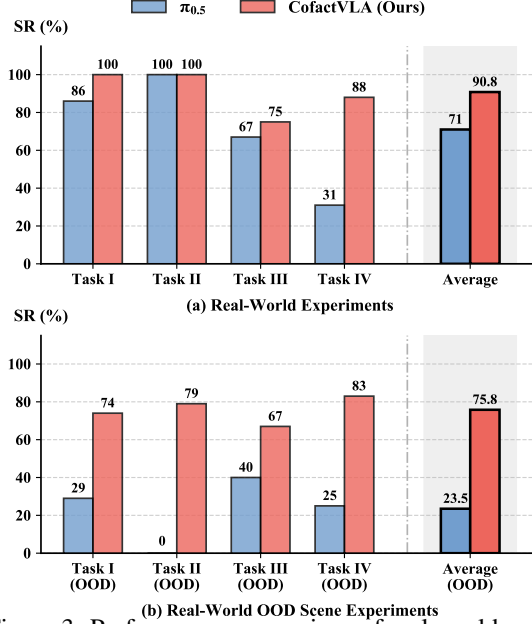


Figure 3: Performance comparison of real-world experiments and the evaluation of OOD scene.

4.4 Ablation Studies

We further conduct additional ablation studies on the four LIBERO suites to thoroughly validate the effectiveness of our CofactVLA model, encompassing both the model architecture and the different counterfactual designs.

Model Architecture. As shown in Table 3, applying either Feature-Level CCR or Action-Level OPG individually improves the baseline average from 97.0% to 97.5% and 98.0%, respectively, validating their standalone effectiveness. Crucially, combining them yields the highest overall performance (98.5%). This confirms our dual-path design is highly synergistic: CCR explicitly cleanses the latent representation to provide a structurally pure semantic intent, which in turn allows OPG to perform a much cleaner and precise geometric projection during action generation.

Superiority of Orthogonal Projection. We compare OPG against traditional scalar-based interventions: naive Addition (94.0%), Subtraction (96.0%), and CAG [40](97.5%). As theoretically argued, direct scalar subtraction (as in CAG) blindly amplifies non-orthogonal noise, risking out-of-distribution (OOD) action flows. In contrast, our OPG mathematically isolates and projects away only the collinear visual bias. This ensures the semantic guidance strictly enhances the causal intent without distorting the valid robotic action manifold, achieving the best performance (98.5%).

Table 3: Ablation studies on **model architecture, counterfactual designs on the LIBERO benchmark.**

Variants	Spatial	Object	Goal	Long	Avg
Model Architecture:					
Baseline	97.0	99.0	96.0	96.0	97.0
w/ CCR	99.0	99.0	98.0	94.0	97.5
w/ OPG	100.0	99.0	98.0	95.0	98.0
Full CofactVLA	99.0	100.0	98.0	97.0	98.5
Counterfactual design:					
Add	98.0	98.0	92.0	88.0	94.0
Sub	97.0	99.0	98.0	90.0	96.0
CAG[40]	98.0	96.0	99.0	97.0	97.5
OPG (Ours)	99.0	100.0	98.0	97.0	98.5

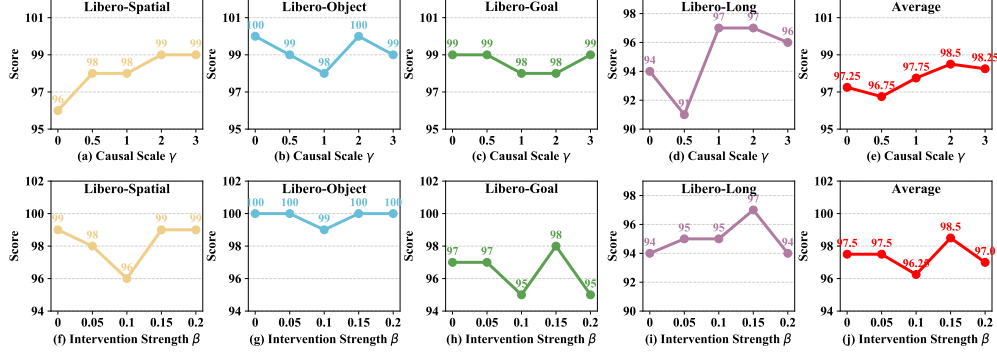


Figure 4: **Sensitivity analyses of intervention intensities.** Top row: action-level causal scale γ . Bottom row: feature-level intervention strength β .

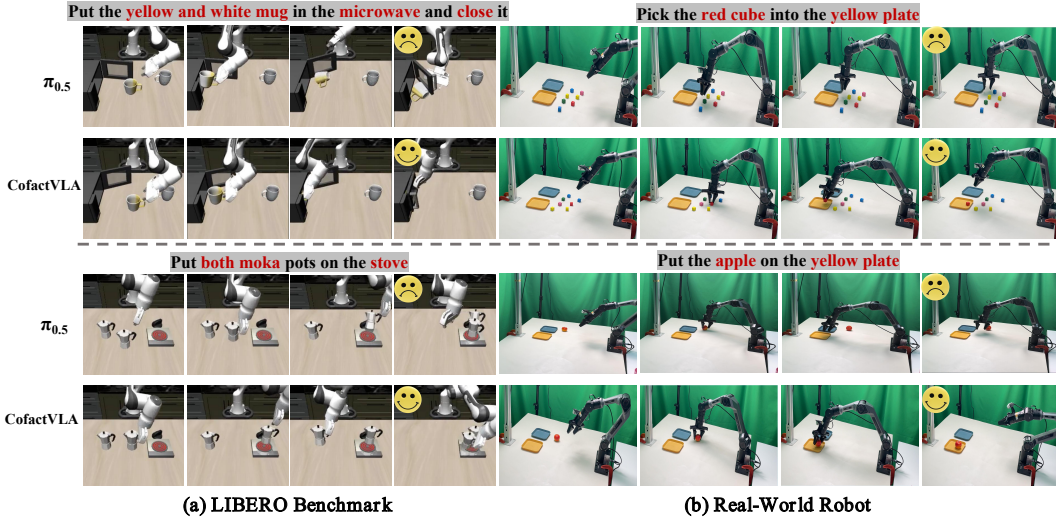


Figure 5: Qualitative comparison of mitigating the ‘vision-override’ phenomenon on the (a) **LIBERO Benchmark** and the (b) **Real-World Robot**.

4.5 Sensitivity Analyses of Intervention Intensity

We investigate the model’s sensitivity to the action-level causal scale γ and the feature-level intervention strength β , as illustrated in Figure 4.

Action-Level Causal Scale (γ): Sweeping $\gamma \in \{0, 0.5, 1, 2, 3\}$ reveals that increasing the guidance scale yields substantial improvements on complex tasks, with Libero-Spatial and Libero-Long peaking at 99% and 97%, respectively. Meanwhile, the performance on Libero-Object and Libero-Goal remains consistently high (98%–100%). However, excessive guidance ($\gamma = 3$) causes minor performance drops. This aligns with continuous generative modeling theory[48, 49]: overly aggressive projection could distort the valid action manifold, causing unnatural trajectories. Thus, we set $\gamma = 2$ for maximizing causal deconfounding and ensuring physical execution safety.

Feature-Level Intervention Strength (β): Similarly, sweeping β across $\{0, 0.05, 0.1, 0.15, 0.2\}$ demonstrates a clear and consistent performance peak at $\beta = 0.15$, achieving the highest average success rate of 98.5%. At this specific threshold, tasks requiring strict semantic grounding and long-horizon planning (i.e., Libero-Goal and Libero-Long) hit their maximum success rates of 98% and 97%, respectively. Conversely, applying an overly aggressive covariance penalty ($\beta = 0.2$) leads to noticeable performance degradation. This indicates that while moderate penalization successfully suppresses visual confounders, excessive feature-level intervention risks inadvertently washing out the essential, language-aligned visual features required for accurate manipulation.

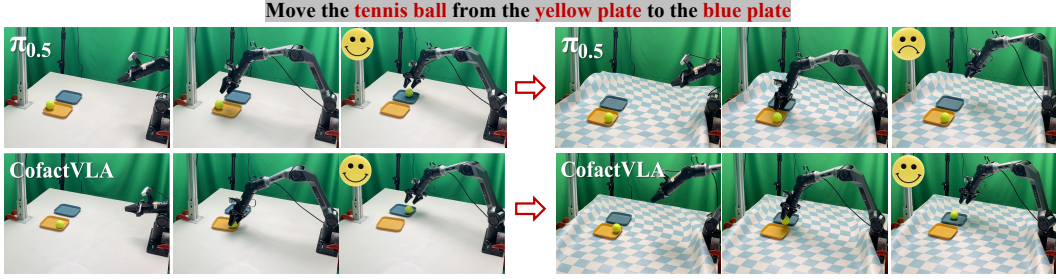


Figure 6: **Qualitative comparison of OOD generalization.** While $\pi_{0.5}$ succeeds on the standard background (left), it fails completely on the unseen checkered texture (right). In contrast, CofactVLA robustly completes the task by decoupling visual confounders.

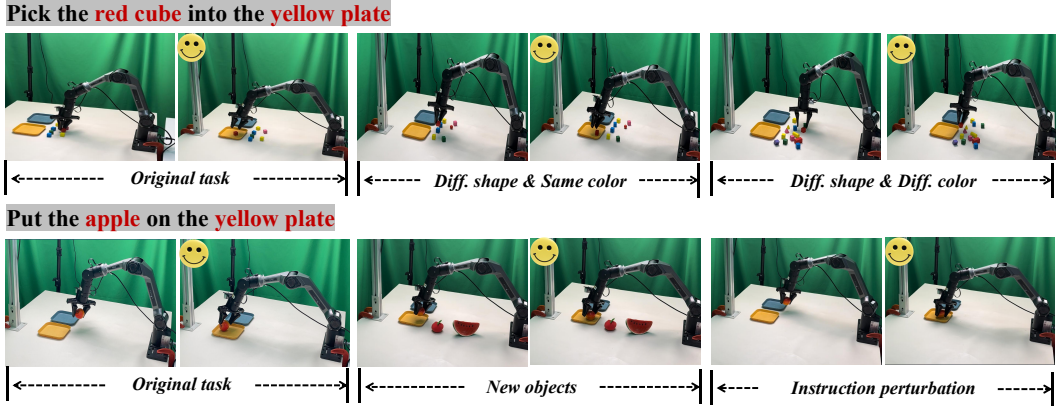


Figure 7: **OOD generalization of CofactVLA:** Visualizations from real-world experiments.

4.6 Qualitative Results and Analyses

We present several qualitative visualizations in both simulation and real-world environments. Please refer to the Appendix A.6 for more comprehensive qualitative results and analyses.

Mitigating Causal Confusion. As shown in Figure 5, the $\pi_{0.5}$ struggles with precise semantic grounding in visually dense scenes. For instance, $\pi_{0.5}$ fails to close the microwave door in the first example of LIBERO, indicating an incomplete understanding of long-horizon linguistic intents. Similarly, in the physical real-world task, $\pi_{0.5}$ fails to accurately locate the target among multiple colorful distractors. By suppressing dominant visual shortcuts, CofactVLA ensures that the generated trajectories are robust, complete, and driven by the causal language instructions.

Generalization to Unseen OOD Environments. Figure 6 highlights our method’s OOD robustness. While both models succeed on standard backgrounds (left), the baseline $\pi_{0.5}$ collapses under an unseen checkered texture (right). Conversely, CofactVLA successfully neutralizes this spurious confounder, maintaining accurate spatial localization. Furthermore, as illustrated in Figure 7, our method extends this zero-shot generalization to object-level and semantic distribution shifts. Even when confronted with novel objects or instruction perturbations, CofactVLA robustly grounds the core semantic intent without overfitting to memorized visual attributes. These results confirm that isolating visual bias directly bridges the generalization gap in diverse and dynamic physical deployments.

5 Conclusion and Limitations

Conclusion. In this paper, we address the “vision-override” phenomenon in VLA models with CofactVLA, a unified causal deconfounding framework based on a Dual-path Deconfounding Graph (DDG). By simultaneously applying Counterfactual Covariance Reduction (CCR) to latent features and Orthogonal Projection Guidance (OPG) to continuous actions, CofactVLA ensures that the generated manipulation trajectories are strictly and consistently grounded in the causal semantic intents of the language instructions. Extensive simulated and real-world experiments demonstrate

state-of-the-art performance, highlighting its exceptional robustness and generalization against dynamic visual distractors in open-world out-of-distribution (OOD) scenarios.

Limitations. CofactVLA is inherently bottlenecked by the base VLM’s zero-shot grounding, as causal interventions cannot synthesize unlearned concepts. Additionally, it remains vulnerable to severe physical occlusions (e.g., the arm obstructing the camera). Future work will explore dynamic multi-view fusion to enhance occlusion-aware robustness.

References

- [1] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Jasmine Hsu, et al. RT-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022.
- [2] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R Sanketi, Grecia Salazar, Michael S Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J Joshi, Alex Irpan, brian ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *7th Annual Conference on Robot Learning*, 2023.
- [3] Xinghang Li, Minghuan Liu, Hanbo Zhang, Cunjun Yu, Jie Xu, Hongtao Wu, Chilam Cheang, Ya Jing, Weinan Zhang, Huaping Liu, Hang Li, and Tao Kong. Vision-language foundation models as effective robot imitators. In *The Twelfth International Conference on Learning Representations*, 2024.
- [4] Hongtao Wu, Ya Jing, Chilam Cheang, Guangzeng Chen, Jiafeng Xu, Xinghang Li, Minghuan Liu, Hang Li, and Tao Kong. Unleashing large-scale video generative pre-training for visual robot manipulation. In *The Twelfth International Conference on Learning Representations*, 2024.
- [5] Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, et al. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025.
- [6] Yi Chen, Yuying Ge, Weiliang Tang, Yizhuo Li, Yixiao Ge, Mingyu Ding, Ying Shan, and Xihui Liu. Moto: Latent motion token as the bridging language for learning robot manipulation from videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 19752–19763, 2025.
- [7] Ruijie Zheng, Yongyuan Liang, Shuaiyi Huang, Jianfeng Gao, Hal Daumé III, Andrey Kolobov, Furong Huang, and Jianwei Yang. TraceVLA: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [8] Shengliang Deng, Mi Yan, Songlin Wei, Haixin Ma, Yuxin Yang, Jiayi Chen, Zhiqi Zhang, Taoyu Yang, Xuheng Zhang, Heming Cui, Zhizheng Zhang, and He Wang. GraspVLA: a grasping foundation model pre-trained on billion-scale synthetic action data. In *Proceedings of The 9th Conference on Robot Learning*, pages 1004–1029. PMLR, 2025.
- [9] Lucy Xiaoyang Shi, brian ichter, Michael Robert Equi, Liyiming Ke, Karl Pertsch, Quan Vuong, James Tanner, Anna Walling, Haohuan Wang, Niccolo Fusai, Adrian Li-Bell, Danny Driess, Lachy Groom, Sergey Levine, and Chelsea Finn. Hi Robot: Open-ended instruction following with hierarchical vision-language-action models. In *Forty-second International Conference on Machine Learning*, 2025.
- [10] Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang, Zhaoye Fei, et al. Libero-plus: In-depth robustness analysis of vision-language-action models. *arXiv preprint arXiv:2510.13626*, 2025.
- [11] Catherine Glossop, William Chen, Arjun Bhorkar, Dhruv Shah, and Sergey Levine. Cast: Counterfactual labels improve instruction following in vision-language-action models. *arXiv preprint arXiv:2508.13446*, 2025.
- [12] Zhenghao Peng, Wenhao Ding, Yurong You, Yuxiao Chen, Wenjie Luo, Thomas Tian, Yulong Cao, Apoorva Sharma, Danfei Xu, Boris Ivanovic, et al. Counterfactual vla: Self-reflective vision-language-action model with adaptive reasoning. *arXiv preprint arXiv:2512.24426*, 2025.

- [13] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.
- [14] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
- [15] Haozhan Li, Yuxin Zuo, Jiale Yu, Yuhao Zhang, Yang Zhaohui, Kaiyan Zhang, Xuekai Zhu, Yuchen Zhang, Tianxing Chen, Ganqu Cui, Dehui Wang, Dingxiang Luo, Yuchen Fan, Youbang Sun, Jia Zeng, Jiangmiao Pang, Shanghang Zhang, Yu Wang, Yao Mu, Bowen Zhou, and Ning Ding. SimpleVLA-RL: Scaling VLA training via reinforcement learning. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [16] Mustafa Shukor, Dana Aubakirova, Francesco Capuano, Pepijn Kooijmans, Steven Palma, Adil Zouitine, Michel Aractingi, Caroline Pascal, Martino Russi, Andres Marafioti, et al. SmolVLA: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025.
- [17] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.
- [18] Moo Jin Kim, Chelsea Finn, and Percy Liang. Fine-tuning vision-language-action models: Optimizing speed and success. *arXiv preprint arXiv:2502.19645*, 2025.
- [19] Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [20] Yihao Wang, Pengxiang Ding, Lingxiao Li, Can Cui, Zirui Ge, Xinyang Tong, Wenxuan Song, Han Zhao, Wei Zhao, Pengxu Hou, et al. VLA-adapter: An effective paradigm for tiny-scale vision-language-action model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 18638–18646, 2026.
- [21] Yan Zhang, Zhong Ji, Yanwei Pang, Jungong Han, and Xuelong Li. Modality-experts coordinated adaptation for large multimodal models. *Science China Information Sciences*, 67(12):220107, 2024.
- [22] Wenxuan Song, Ziyang Zhou, Han Zhao, Jiayi Chen, Pengxiang Ding, Haodong Yan, Yuxin Huang, Feilong Tang, Donglin Wang, and Haoang Li. Reconvla: Reconstructive vision-language-action model as effective robot perceiver. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 18549–18557, 2026.
- [23] Danny Driess, Jost Tobias Springenberg, brian ichter, LILI YU, Adrian Li-Bell, Karl Pertsch, Allen Z. Ren, Homer Walke, Quan Vuong, Lucy Xiaoyang Shi, and Sergey Levine. Knowledge insulating vision-language-action models: Train fast, run fast, generalize better. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [24] Rongyu Zhang, Menghang Dong, Yuan Zhang, Liang Heng, Xiaowei Chi, Gaole Dai, Li Du, Dan Wang, Yuan Du, and Shanghang Zhang. Mole-vla: Dynamic layer-skipping vision language action model via mixture-of-layers for efficient robot manipulation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 18764–18772, 2026.
- [25] Songsheng Wang, Rucheng Yu, Zhihang Yuan, Chao Yu, Feng Gao, Yu Wang, and Derek F Wong. Spec-vla: speculative decoding for vision-language-action models with relaxed acceptance. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 26916–26928, 2025.
- [26] Wei Li, Renshan Zhang, Rui Shao, Jie He, and Liqiang Nie. CogVLA: Cognition-aligned vision-language-action models via instruction-driven routing & sparsification. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [27] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, pages 6840–6851, 2020.
- [28] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44(10-11):1684–1704, 2025.
- [29] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4195–4205, 2023.

- [30] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *The Eleventh International Conference on Learning Representations*, 2023.
- [31] Xingchao Liu, Chengyue Gong, and qiang liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *The Eleventh International Conference on Learning Representations*, 2023.
- [32] Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. RDT-1b: a diffusion foundation model for bimanual manipulation. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [33] Zhendong Wang, Max Li, Ajay Mandlekar, Zhenjia Xu, Jiaojiao Fan, Yashraj Narang, Linxi Fan, Yuke Zhu, Yogesh Balaji, Mingyuan Zhou, Ming-Yu Liu, and Yu Zeng. One-step diffusion policy: Fast visuomotor policies via diffusion distillation. In *Forty-second International Conference on Machine Learning*, 2025.
- [34] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- [35] Jiaming Liu, Hao Chen, Pengju An, Zhuoyang Liu, Renrui Zhang, Chenyang Gu, Xiaoqi Li, Ziyu Guo, Sixiang Chen, Mengzhen Liu, et al. Hybridvla: Collaborative diffusion and autoregression in a unified vision-language-action model. *arXiv preprint arXiv:2503.10631*, 2025.
- [36] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.
- [37] Jonathan Richens and Tom Everitt. Robust agents learn causal world models. In *The Twelfth International Conference on Learning Representations*, 2024.
- [38] Ossama Ahmed, Frederik Träuble, Anirudh Goyal, Alexander Neitz, Manuel Wuthrich, Yoshua Bengio, Bernhard Schölkopf, and Stefan Bauer. Causalworld: A robotic manipulation benchmark for causal structure and transfer learning. In *International Conference on Learning Representations*, 2021.
- [39] Jiahua Ma, Yiran Qin, Yixiong Li, Xuanqi Liao, Yulan Guo, and Ruimao Zhang. CDP: Towards robust autoregressive visuomotor policy learning via causal diffusion. In *Proceedings of the 9th Conference on Robot Learning (CoRL)*, pages 4190–4205. PMLR, 2025.
- [40] Yu Fang, Yuchun Feng, Dong Jing, Jiaqi Liu, Yue Yang, Zhenyu Wei, Daniel Szafrir, and Mingyu Ding. When vision overrides language: Evaluating and mitigating counterfactual failures in vlcs. *arXiv preprint arXiv:2602.17659*, 2026.
- [41] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. Libero: Benchmarking knowledge transfer for lifelong robot learning. In *The Thirty-seventh Annual Conference on Neural Information Processing Systems*, 2023.
- [42] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, XinQiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [43] Jinliang Zheng, Jianxiong Li, Zhihao Wang, Dongxiu Liu, Xirui Kang, Yuchun Feng, Yinan Zheng, Jiayin Zou, Yilun Chen, Jia Zeng, Tai Wang, Ya-Qin Zhang, Jingjing Liu, and Xianyu Zhan. X-VLA: Soft-prompted transformer as scalable cross-embodiment vision-language-action model. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [44] Chia-Yu Hung, Qi Sun, Pengfei Hong, Amir Zadeh, Chuan Li, U Tan, Navonil Majumder, Soujanya Poria, et al. NORA: A small open-sourced generalist vision language action model for embodied tasks. *arXiv preprint arXiv:2504.19854*, 2025.
- [45] Jun Cen, Chaohui Yu, Hangjie Yuan, Yuming Jiang, Siteng Huang, Jiayan Guo, Xin Li, Yibing Song, Hao Luo, Fan Wang, et al. Worldvla: Towards autoregressive action world model. *arXiv preprint arXiv:2506.21539*, 2025.
- [46] Qingwen Bu, Yanting Yang, Jisong Cai, Shenyuan Gao, Guanghui Ren, Maoqing Yao, Ping Luo, and Hongyang Li. UniVLA: Learning to act anywhere with task-centric latent actions. *arXiv preprint arXiv:2505.06111*, 2025.

- [47] Karl Pertsch, Kyle Stachowicz, Brian Ichter, Danny Driess, Suraj Nair, Quan Vuong, Oier Mees, Chelsea Finn, and Sergey Levine. Fast: Efficient action tokenization for vision-language-action models. *arXiv preprint arXiv:2501.09747*, 2025.
- [48] Jinhao Liang, Yixuan Sun, Anirban Samaddar, Sandeep Madireddy, and Ferdinando Fioretto. Chance-constrained flow matching for high-fidelity constraint-aware generation. *arXiv preprint arXiv:2509.25157*, 2025.
- [49] Jean-Baptiste Bouvier, Kanghyun Ryu, Kartik Nagpal, Qiayuan Liao, Koushil Sreenath, and Negar Mehr. Ddat: Diffusion policies enforcing dynamically admissible robot trajectories. *arXiv preprint arXiv:2502.15043*, 2025.
- [50] Remi Cadene, Simon Alibert, Francesco Capuano, Michel Aractingi, Adil Zouitine, Pepijn Kooijmans, Jade Choghari, Martino Russi, Caroline Pascal, Steven Palma, Dana Aubakirova, Mustafa Shukor, Jess Moss, Alexander Soare, Quentin Lhoest, Quentin Gallouédec, and Thomas Wolf. LeRobot: An open-source library for end-to-end robot learning. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [51] Xueyang Zhou, Yangming Xu, Guiyao Tie, Yongchao Chen, Guowen Zhang, Duanfeng Chu, Pan Zhou, and Lichao Sun. Libero-pro: Towards robust and fair evaluation of vision-language-action models beyond memorization. *arXiv preprint arXiv:2510.03827*, 2025.

A Technical appendices and supplementary material

In this appendix, we further provide the following additional details, which are omitted in the main paper owing to the limited space.

The structure of this Appendix is as follows:

- Section A.1 contains **additional methodological proofs**.
- Section A.2 contains **introduction of our real-world robot platform**.
- Section A.3 contains **training configuration and implementation details**.
- Section A.4 contains **details of perturbation experiment design**.
- Section A.5 contains **additional real-world robot experiments**.
- Section A.6 contains **comprehensive qualitative results and analyses**.
- Section A.7 contains **limitations and outlooks**.
- Section A.8 contains **broader impacts**.

A.1 Additional Methodological Proofs

A.1.1 Preliminaries: Whitening and Generalized Eigenvectors

Throughout, assume $\Sigma_0 \succ 0$. Define the Σ_0 -whitened matrix

$$M := \Sigma_0^{-1/2} \Sigma_\Delta \Sigma_0^{-1/2}, \quad (10)$$

which is symmetric positive semidefinite. A pair (λ, b) solves the generalized eigenvalue problem

$$\Sigma_\Delta b = \lambda \Sigma_0 b. \quad (11)$$

If and only if (λ, u) solves the standard eigenproblem

$$Mu = \lambda u, \quad u := \Sigma_0^{1/2} b. \quad (12)$$

Indeed, multiplying (11) on the left by $\Sigma_0^{-1/2}$ and setting $u = \Sigma_0^{1/2} b$ gives (12); conversely, $b = \Sigma_0^{-1/2} u$ maps eigenvectors of M to generalized eigenvectors of $(\Sigma_\Delta, \Sigma_0)$.

A.1.2 Closed-form Solution of the Generalized Eigenproblem

Proposition 1 Consider the optimization problem: $\max_{B \in \mathbb{R}^{d_r \times d_v}} \text{tr}(B^\top \Sigma_\Delta B)$ subject to $B^\top \Sigma_0 B = I_{d_v}$. Any optimizer B^* for this problem spans the top d_v generalized eigenspace of $(\Sigma_\Delta, \Sigma_0)$:

$$\Sigma_\Delta b_i = \lambda_i \Sigma_0 b_i, \quad \lambda_1 \geq \dots \geq \lambda_{d_r} \geq 0, \quad B^* = [b_1, \dots, b_{d_v}]. \quad (13)$$

Proof 1 Let $C := \Sigma_0^{1/2} B \in \mathbb{R}^{d_r \times d_v}$. Then the constraint becomes

$$B^\top \Sigma_0 B = I_{d_v} \iff (\Sigma_0^{1/2} B)^\top (\Sigma_0^{1/2} B) = C^\top C = I_{d_v}. \quad (14)$$

The objective becomes

$$\text{tr}(B^\top \Sigma_\Delta B) = \text{tr}(B^\top \Sigma_0^{1/2} \underbrace{\Sigma_0^{-1/2} \Sigma_\Delta \Sigma_0^{-1/2}}_M \Sigma_0^{1/2} B) = \text{tr}(C^\top M C). \quad (15)$$

Let $M = Q \Lambda Q^\top$ be an eigendecomposition with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{d_r} \geq 0$ and orthonormal eigenvectors $Q = [q_1, \dots, q_{d_r}]$. By the Ky Fan maximum principle (a trace form of Rayleigh–Ritz), the maximum attained by

$$C^* = [q_1, \dots, q_{d_v}] U, \quad (16)$$

for any orthogonal $U \in \mathbb{R}^{d_v \times d_v}$ (i.e., any orthonormal basis of the top- d_v eigenspace). Mapping back yields

$$B^* = \Sigma_0^{-1/2} C^*. \quad (17)$$

Each column $b_i = \Sigma_0^{-1/2} q_i$ satisfies the generalized eigen-equation:

$$\Sigma_\Delta b_i = \Sigma_\Delta \Sigma_0^{-1/2} q_i = \Sigma_0^{1/2} (\Sigma_0^{-1/2} \Sigma_\Delta \Sigma_0^{-1/2}) q_i = \Sigma_0^{1/2} M q_i = \lambda_i \Sigma_0^{1/2} q_i = \lambda_i \Sigma_0 b_i. \quad (18)$$

Hence any optimizer spans the top d_v generalized eigenspace of $(\Sigma_\Delta, \Sigma_0)$, proving (13).

A.1.3 Π_V is the Σ_0 -orthogonal Projector

Lemma 1 Let $B \in \mathbb{R}^{d_r \times d_v}$ have full column rank and define

$$\Pi_V := B(B^\top \Sigma_0 B)^{-1} B^\top \Sigma_0. \quad (19)$$

Then (i) $\Pi_V^2 = \Pi_V$ (idempotent), (ii) $\text{range}(\Pi_V) = \text{span}(B)$, and (iii) Π_V is self-adjoint w.r.t. the Σ_0 -inner product:

$$\langle u, \Pi_V v \rangle_{\Sigma_0} = \langle \Pi_V u, v \rangle_{\Sigma_0}, \quad \langle a, b \rangle_{\Sigma_0} := a^\top \Sigma_0 b. \quad (20)$$

Consequently, Π_V is the Σ_0 -orthogonal projector onto $\text{span}(B)$ and $\Pi_\perp := I - \Pi_V$ projects onto the Σ_0 -orthogonal complement.

Proof 2 Let $G := B^\top \Sigma_0 B \in \mathbb{R}^{d_v \times d_v}$; G is invertible since $\Sigma_0 \succ 0$ and B has full column rank. Then

$$\Pi_V^2 = BG^{-1} B^\top \Sigma_0 BG^{-1} B^\top \Sigma_0 = BG^{-1} \underbrace{(B^\top \Sigma_0 B)}_G G^{-1} B^\top \Sigma_0 = BG^{-1} B^\top \Sigma_0 = \Pi_V. \quad (21)$$

Thus Π_V is idempotent. Also, for any vector z , $\Pi_V z \in \text{span}(B)$ by construction, so $\text{range}(\Pi_V) \subseteq \text{span}(B)$. Conversely, for any $y = Bc$,

$$\Pi_V(Bc) = BG^{-1} B^\top \Sigma_0 Bc = BG^{-1} Gc = Bc = y, \quad (22)$$

so $\text{span}(B) \subseteq \text{range}(\Pi_V)$. Hence $\text{range}(\Pi_V) = \text{span}(B)$.

For self-adjointness, note $\Pi_V^\top \Sigma_0 = \Sigma_0 \Pi_V$:

$$\Pi_V^\top \Sigma_0 = (\Sigma_0^\top B G^{-1} B^\top) \Sigma_0 = \Sigma_0 B G^{-1} B^\top \Sigma_0 = \Sigma_0 \Pi_V, \quad (23)$$

where we used $\Sigma_0^\top = \Sigma_0$. Therefore for any u, v ,

$$\langle u, \Pi_V v \rangle_{\Sigma_0} = u^\top \Sigma_0 \Pi_V v = u^\top \Pi_V^\top \Sigma_0 v = (\Pi_V u)^\top \Sigma_0 v = \langle \Pi_V u, v \rangle_{\Sigma_0}. \quad (24)$$

This is exactly Σ_0 -orthogonal projection onto $\text{span}(B)$.

A.1.4 CCR: Delete Nuisance Bias, Keep Content Gain

We proceed in three steps.

Step 1: Top generalized eigenspace equals top eigenspace of M . By the whitening equivalence (12), generalized eigenvectors of $(\Sigma_\Delta, \Sigma_0)$ correspond to eigenvectors of $M = \Sigma_0^{-1/2} \Sigma_\Delta \Sigma_0^{-1/2}$ under the invertible map $u = \Sigma_0^{1/2} b$.

Step 2: Subspace separation implies the top- d_v eigenspace is $\Sigma_0^{1/2} \mathcal{S}_O$. Let

$$\mathcal{T}_O := \Sigma_0^{1/2} \mathcal{S}_O, \quad \mathcal{T}_C := \Sigma_0^{1/2} \mathcal{S}_C. \quad (25)$$

Since $\Sigma_0^{1/2}$ is invertible and $\mathbb{R}^{d_r} = \mathcal{S}_C \oplus \mathcal{S}_O$, we also have $\mathbb{R}^{d_r} = \mathcal{T}_O \oplus \mathcal{T}_C$ and $\dim(\mathcal{T}_O) = d_o$.

Assumption 2 states that eigenvalues of M on $\mathcal{T}_C$ are strictly larger than those on $\mathcal{T}_O$:

$$\lambda_{\min}(M|_{\mathcal{T}_C}) > \lambda_{\max}(M|_{\mathcal{T}_O}). \quad (26)$$

This implies a spectral separation: the top d_c (let $\dim(\mathcal{T}_C) = d_c$) eigenvalues of M come from the restriction to $\mathcal{T}_C$, and therefore the top- d_c eigenspace of M equals $\mathcal{T}_C$. Concretely, if U_{bias} is any orthonormal basis of $\mathcal{T}_C$, then U_{bias} spans the maximizer and thus spans the top- d_o eigenspace of M (uniquely up to rotations within that eigenspace).

Mapping back via $b = \Sigma_0^{-1/2} u$, the top generalized eigenspace equals $\mathcal{S}_C$. Hence the matrix B formed by the top generalized eigenvectors satisfies $\text{span}(B) = \mathcal{S}_C$.

Step 3: $\Pi_{\perp}$ removes exactly the nuisance component of $\tilde{\Delta}$. By Lemma 1, let Π_C be the Σ_0 -orthogonal projector onto $\text{span}(B) = \mathcal{S}_C$, and $\Pi_{\perp} = I - \Pi_C$ projects onto the Σ_0 -orthogonal complement, which is $\mathcal{S}_O$. Therefore, $\Pi_{\perp}\tilde{\Delta} = (I - \Pi_C)(\tilde{\Delta}_C + \tilde{\Delta}_O) = \tilde{\Delta}_O$. This isolated $\tilde{\Delta}_O$ allows us to extract the causal semantic features.

Finally, with $\Delta = \Delta_C + \Delta_O$ (the non-token version of the same decomposition),

$$F_{\text{causal}} = F + \Delta_O = F - \beta(FU_{\text{bias}})U_{\text{bias}}^{\top}, \quad (27)$$

which is exactly (9).

A.2 Introduction of our Real-world Robot Platform

Hardware Platform. As illustrated in Figure 8, our physical experiments are conducted using a 6-DoF AgileX PiPer robotic arm, which provides precise and dexterous manipulation capabilities. The end-effector is a standard 1-DoF parallel gripper. To perceive the environment and provide visual feedback for the continuous flow matching policy, we deploy two Intel RealSense D435 cameras. One camera is mounted overhead to capture a global scene view (*Third Person View*), while the other is attached to the robot’s wrist to capture fine-grained, localized spatial details (*Wrist View*) during manipulation. The diverse set of objects used in our experiments, including various fruits, vegetables, blocks, and plates, is shown in Figure 8(b).

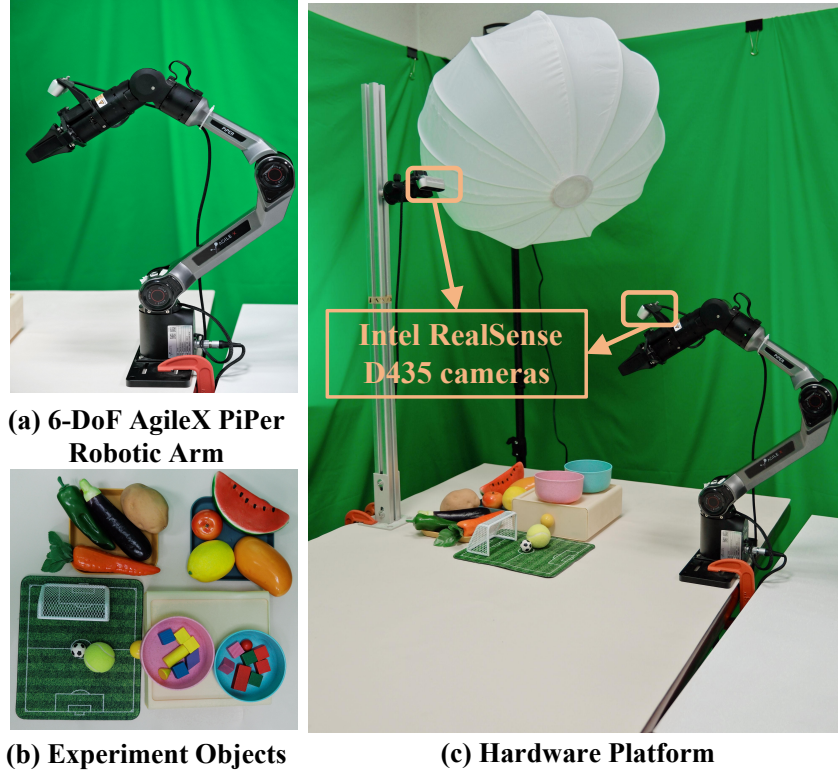


Figure 8: Our hardware platform for the real-world robot experiments.

A.3 Training Configuration and Implementation Details

We build CofactVLA upon the Hugging Face 1erobot [50] framework, where the core architecture is initialized from the pre-trained $\pi_{0.5}$ [17] checkpoint. As summarized in Table 4, all model fine-tuning is conducted on a computing node equipped with 4×NVIDIA H100 (96GB) GPUs. Detailed hyperparameters are summarized in Table 4.

Datasets and Evaluation Protocols. For **simulation**, our model is trained exclusively on the standard LIBERO dataset [41], which encompasses four benchmark suites: **LIBERO-Spatial**, **LIBERO-Object**, **LIBERO-Goal**, and **LIBERO-Long** (or LIBERO-10). These suites focus on various spatial

relationships, object categories, goal objectives, and extended sequential challenges, respectively. We evaluate these standard suites over 10 episodes to ensure statistical reliability. To assess zero-shot robustness, we directly deploy the trained policy on the LIBERO-Plus [10] benchmarks. Due to the extensive scale of LIBERO-Plus, its tasks are evaluated for a single episode. For **real-world deployments**, we collect a custom dataset comprising approximately 400 expert trajectories (about 100 per task). Real-world evaluations report the average success rate over 100 independent trials per task. We provide several trajectories in Figures 9-10.

Asset Licenses. Our implementation relies on several open-source assets. The LIBERO benchmark suites are licensed under the MIT License. The 1erobot[50] framework and the pre-trained $\pi_{0.5}$ checkpoints are distributed under the Apache License 2.0. Our usage of these datasets and models strictly adheres to their respective terms of use.

Table 4: CofactVLA Training Configuration.

Hyperparameters	Value	Hyperparameters	Value
Compute resources	4×H100-96GB	Causal scale γ	2.0
Backbone	$\pi_{0.5}$ [17]	Intervention strength β	0.15
Batch size	32 / GPU	Intervention layers	[15, 16]
Learning rate (LR)	2.5e-5	Freeze vision encoder	False
Training steps	6K	Freeze action expert	False
Weight decay	0.01	Gradient checkpointing	True
Optimizer	AdamW	Action chunk size	50
Betas	[0.9, 0.95]	Warm-up steps	1K

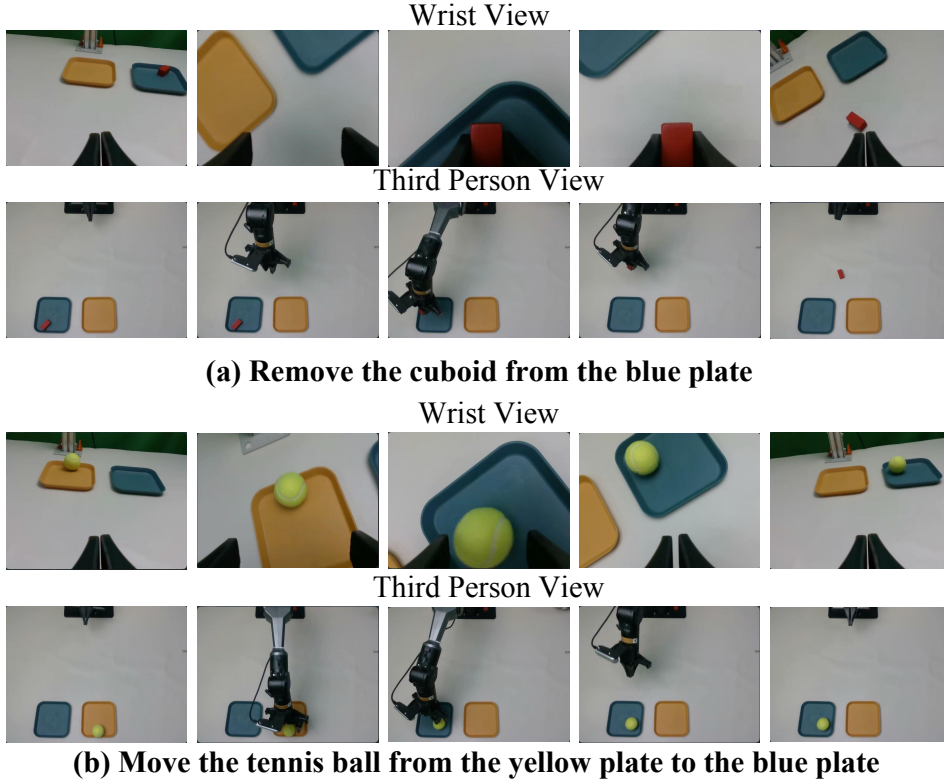


Figure 9: Example of dual-view sample collection for Task I and Task II.

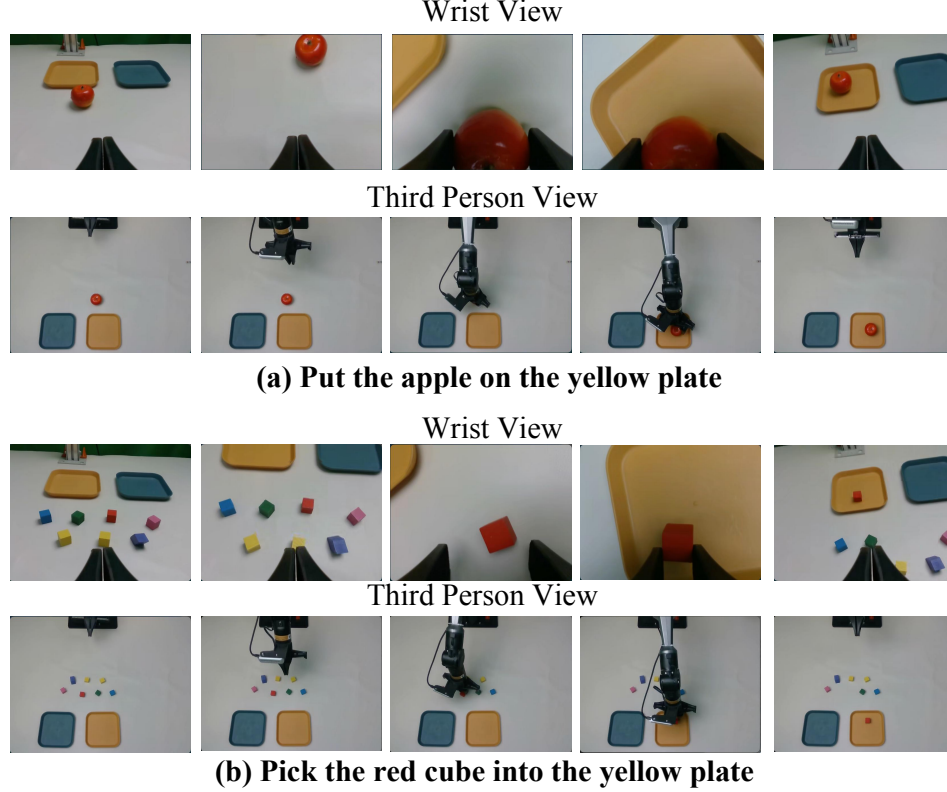


Figure 10: Example of dual-view sample collection for Task III and Task IV.

A.4 Perturbation Experiment Design

Inspired by the task design of LIBERO-Pro [51], we delicately design several comprehensive perturbation experiments to assess anti-perturbation ability of the VLA models beyond the visual layouts and object configurations. In this section, we introduce perturbations along three dimensions, including *object perturbation*, *environment perturbations*, and *instruction perturbations*, to reveal whether the VLA models rely on genuine perception and language grounding or merely memorize task-specific trajectories. Figure 11 provides an intuitive demonstration, and the specific design of perturbation for different tasks are as follows:

- For *Task I*: *remove the cuboid from the blue plate*, we introduce both *object perturbation* and *environment perturbation*. For object-level perturbation, several objects with different shapes and colors are placed around the target red cuboid, making the target less visually salient and increasing the difficulty of object detection and grounding for the VLA models. For environment-level perturbation, the original white tabletop is covered with a green grass-like surface, introducing a significant background texture shift. This setting evaluates whether the model could still accurately localize the target under foreground distractors or background appearance changes.
- For *Task II*: *move the tennis ball from the yellow plate to the blue plate*, we further introduce more challenging *environment perturbations* to test the robustness of the learned policy. Specifically, the original plates are replaced with lighter-colored yellow plates, and the tabletop is wholly covered with a blue checkered tablecloth. These modifications change the surroundings of both the source and the target workspace. This perturbation is designed to examine whether the model could maintain reliable manipulation when the environment distribution differs from the training data.
- For *Task III*: *put the apple on the yellow plate*, we apply both *object perturbation* and *instruction perturbation*. For object-level perturbation, we place visually similar distractors near the target apple, including a red tomato and a piece of watermelon. Since these

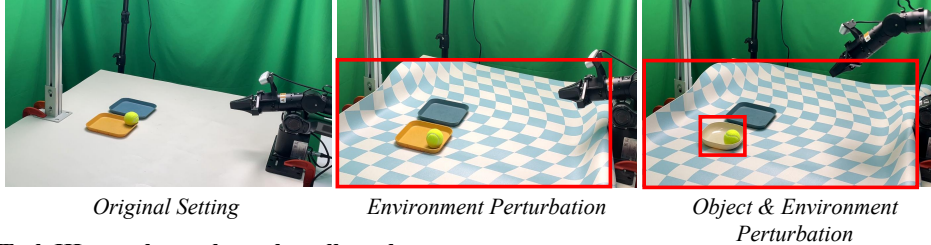
objects share similar category-level or color-level cues with the red apple, they inevitably interfere with fine-grained object recognition of the VLA models. In addition, we introduce instruction-level perturbation by paraphrasing the original prompt, such as replacing the expression “put ... on” with “pick ... into”. This setting tests whether the model could jointly handle visual distractors and linguistic variation.

- For *Task IV: pick the red cube into the yellow plate*, we design a more cluttered and fine-grained object perturbation scenario. The original set of seven small wooden blocks is expanded to eight objects, and multiple blocks are randomly scattered across the workspace. For example, we place a red sphere whose appearance is highly similar to the target red cube in color and size but differs in shape. This setting aims to evaluate the ability of correctly grounding the compositional description “red cube” by jointly considering both color and shape, rather than relying only on the most salient color cue.

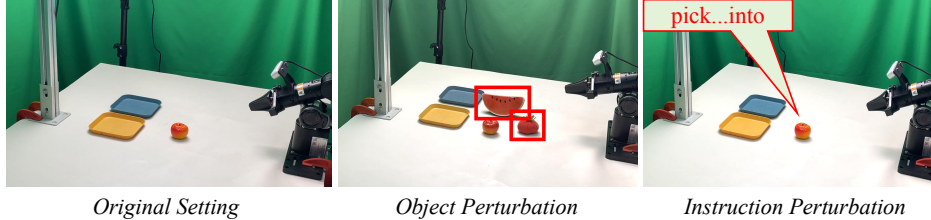
Task I: remove the cuboid from the blue plate



Task II: move the tennis ball from the yellow plate to the blue plate



Task III: put the apple on the yellow plate



Task IV: pick the red cube into the yellow plate

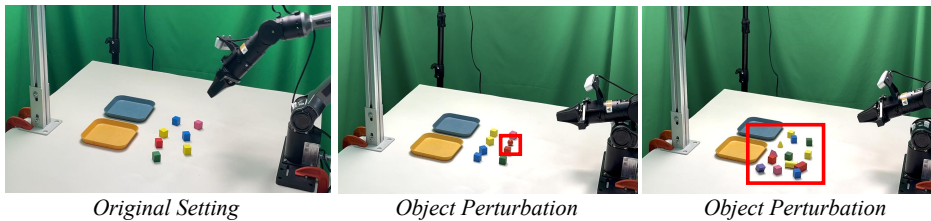


Figure 11: Visualizations of the perturbation design for different tasks.

A.5 Additional Real-world Robot Experiments

In this section, we provide additional real-world robot experiments to validate the robustness and generalization of our method. As it is shown in Table 5, we evaluate real-robot manipulation on these tasks and compare against OpenVLA[14], π_0 [34], and $\pi_{0.5}$ [17]. The tasks span a range of manipulation primitives, including: (i) object placement into receptacles (apple→plate, tennis ball→bowl,

Table 5: Task IDs and their instructions.

Task ID	Instructions
A-1	Pick the apple into plate
A-2	Pick the tennis into bowl
A-3	Place the eggplant into the bowl
A-4	Place the tissue into the plate
A-5	Stack the bag on the plate
A-6	Open the drawer
A-7	Close the drawer
A-8	Open the drawer and pick up the apple
A-9	Pick the pen and insert into cup

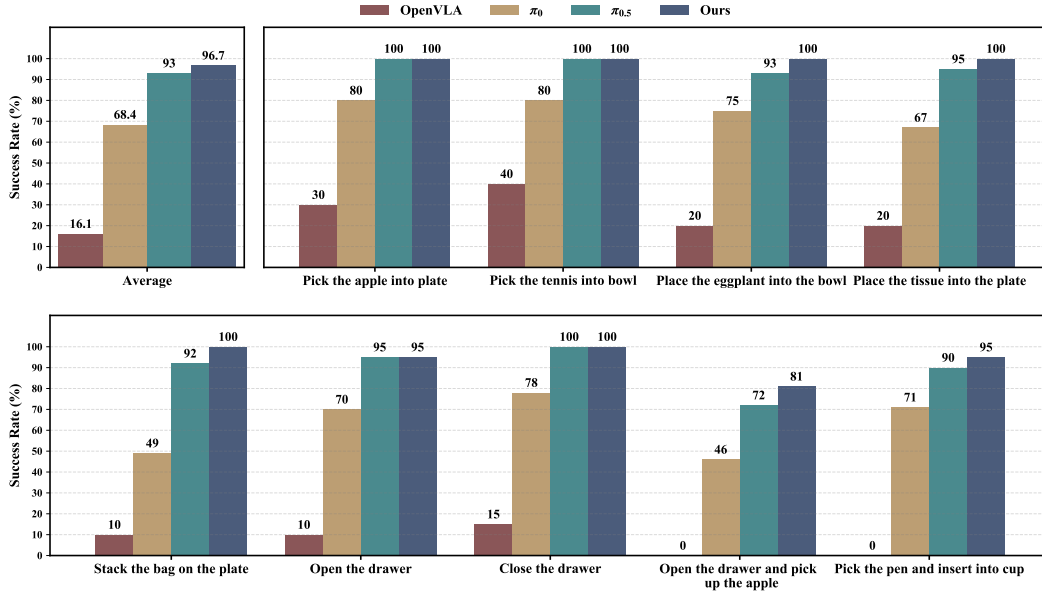


Figure 12: We report success rate (100 trials) for 9 skills (x-axis) and the average over all tasks (left inset), comparing OpenVLA, π_0 , $\pi_{0.5}$, and Ours. Our method achieves the best overall performance (96.7% average) and reaches better performance on most tasks.

eggplant→bowl, tissue→plate), (ii) stacking/placing object onto another surface (bag→plate), (iii) push and pull (opening and closing a drawer), (iv) longer-horizon sequential tasks (open the drawer → pick up the apple, insert a pen into a cup).

All methods are evaluated under the same real-robot setup and identical execution protocol. Figure 12 summarizes success rates, and our method achieves the best average performance (**96.7%**), improving over the strongest baseline $\pi_{0.5}$ (93.0%) by **+3.7** points and clearly outperforming π_0 (68.4%) and OpenVLA (16.1%). While both our method and $\pi_{0.5}$ saturate on simple pick-and-place tasks (100%), our gains concentrate on different object manipulation (e.g., tissue placement and bag stacking, up to **+8** over $\pi_{0.5}$), as well as long-horizon compositions (e.g., Open drawer + pick and Pen insertion).

A.6 Comprehensive Qualitative Results and Analyses

To provide a deeper understanding of our proposed CofactVLA framework, we present extensive qualitative visualizations across both simulated benchmarks and real-world physical deployments. These examples explicitly demonstrate our method’s superiority in semantic grounding, robustness against distribution shifts, and current operational boundaries.

Comparisons against Baselines on LIBERO. Figure 13 provides a detailed step-by-step comparison between the baseline $\pi_{0.5}$ [17] and our CofactVLA in the standard LIBERO simulation environment. As illustrated, while $\pi_{0.5}$ frequently suffers from causal confusion by prematurely interacting with

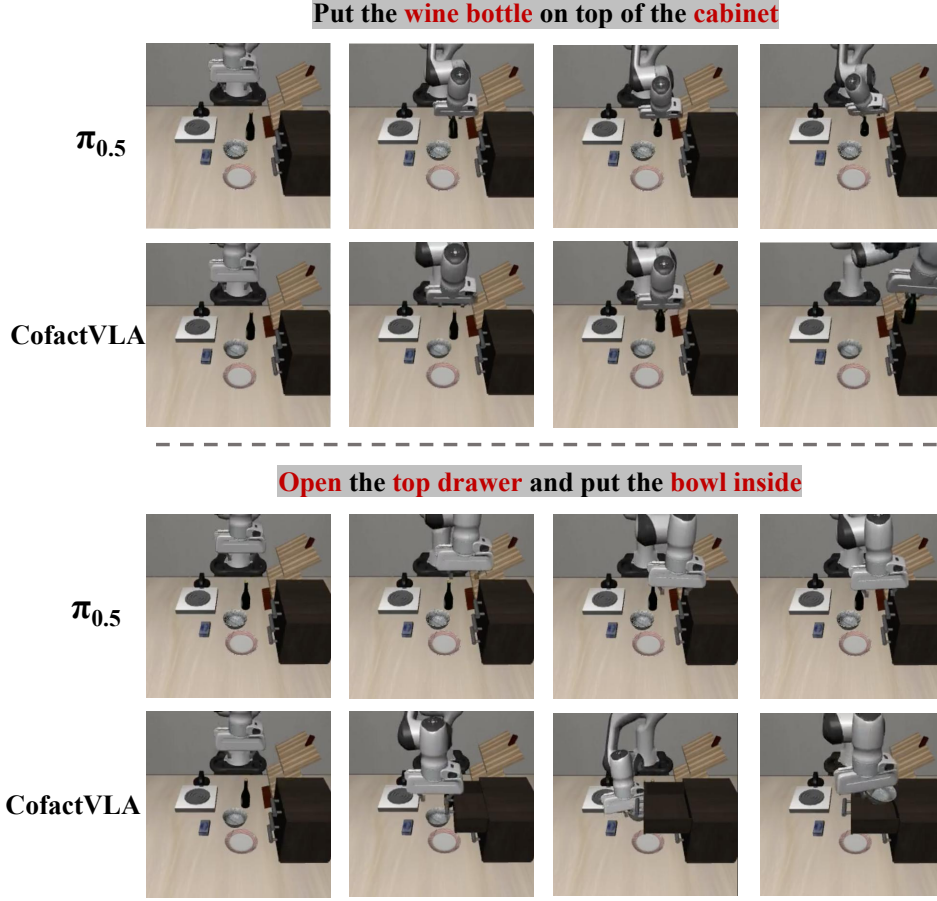


Figure 13: Qualitative comparison between our CofactVLA model and $\pi_{0.5}$ model on the LIBERO Benchmark.

visually prominent but task-irrelevant objects, CofactVLA successfully maintains strict alignment with the language instruction. This visual evidence further corroborates that our dual-path deconfounding mechanism effectively suppresses spurious visual shortcuts.

Robustness Across the LIBERO-Plus Suites. To validate the out-of-distribution (OOD) robustness of our approach, Figures 14 to 17 showcase successful execution trajectories on the highly challenging LIBERO-Plus dataset. We categorize these visualizations into four distinct suites: *Spatial*, *Object*, *Goal*, and *Long*. Despite the introduction of complex visual perturbations (e.g., altered lighting, novel camera viewpoints, and unseen textures), CofactVLA exhibits remarkable zero-shot generalization. Notably, in the *Long* suite (Figure 17), our policy robustly completes extended sequential tasks without suffering from the compounding errors typically induced by visual confounders.

Real-World Deployment and Generalization. Figure 18 extends our qualitative analysis to physical environments, including the standard execution of the original tasks and additional generalization scenarios as introduced in Appendix A.4. For instance, when presented with entirely unseen table textures or dynamically added physical distractors, CofactVLA dynamically neutralizes these visual biases. The robotic arm smoothly reaches the target receptacles without hesitation, bridging the notoriously difficult sim-to-real and environment-to-environment generalization gaps.

Failure Mode Analysis. Finally, to provide a transparent and rigorous evaluation of our system, we present representative failure cases from both LIBERO-Plus and real-world experiments in Figure 19 and Figure 20, respectively. In the simulation environment, failures primarily occur when multiple perturbations are overlaid simultaneously (e.g., extreme lighting changes (the 4th instance in Figure 19) combined with drastic camera angle shifts), occasionally causing the vision encoder to lose spatial tracking. In the real-world deployments, execution failures are predominantly observed under severe

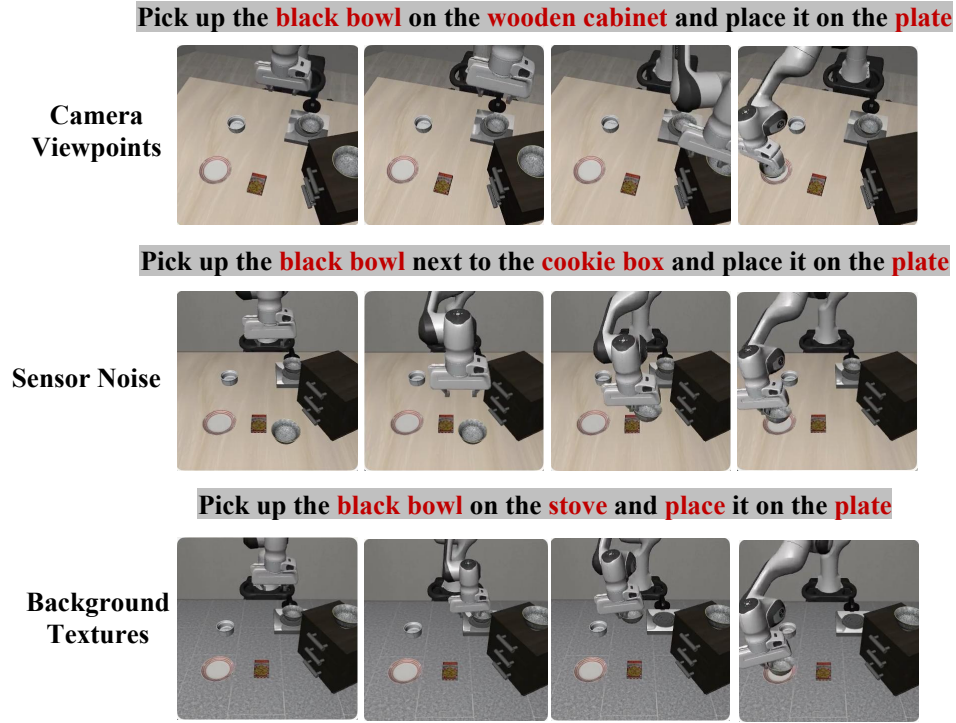


Figure 14: Qualitative results of our CofactVLA model on the **Spatial suite of LIBERO-Plus Benchmark**.

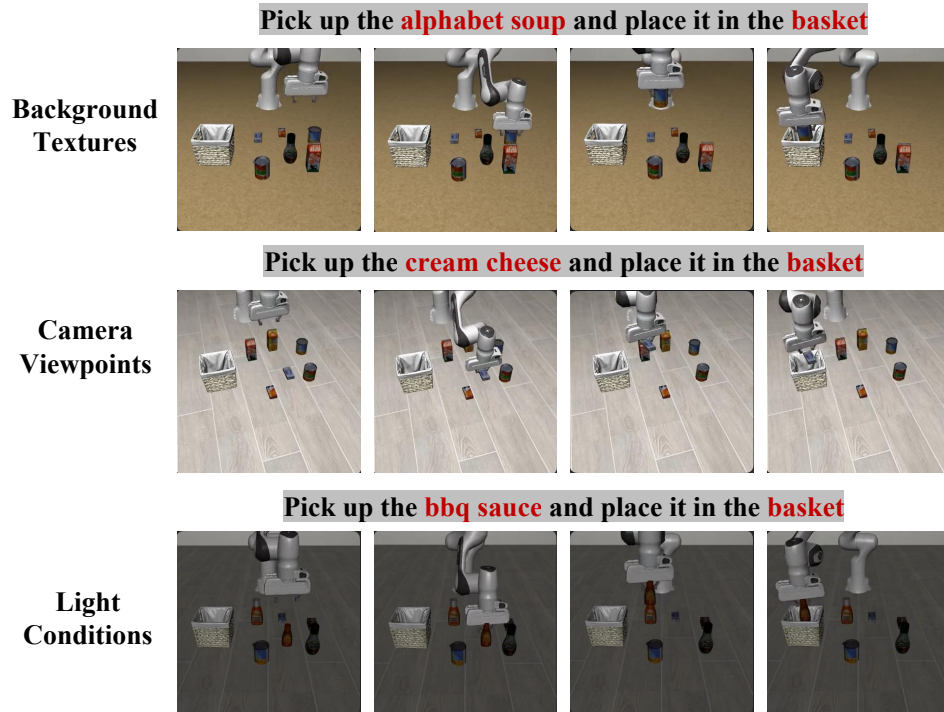


Figure 15: Qualitative results of our CofactVLA model on the **Object suite of LIBERO-Plus Benchmark**.

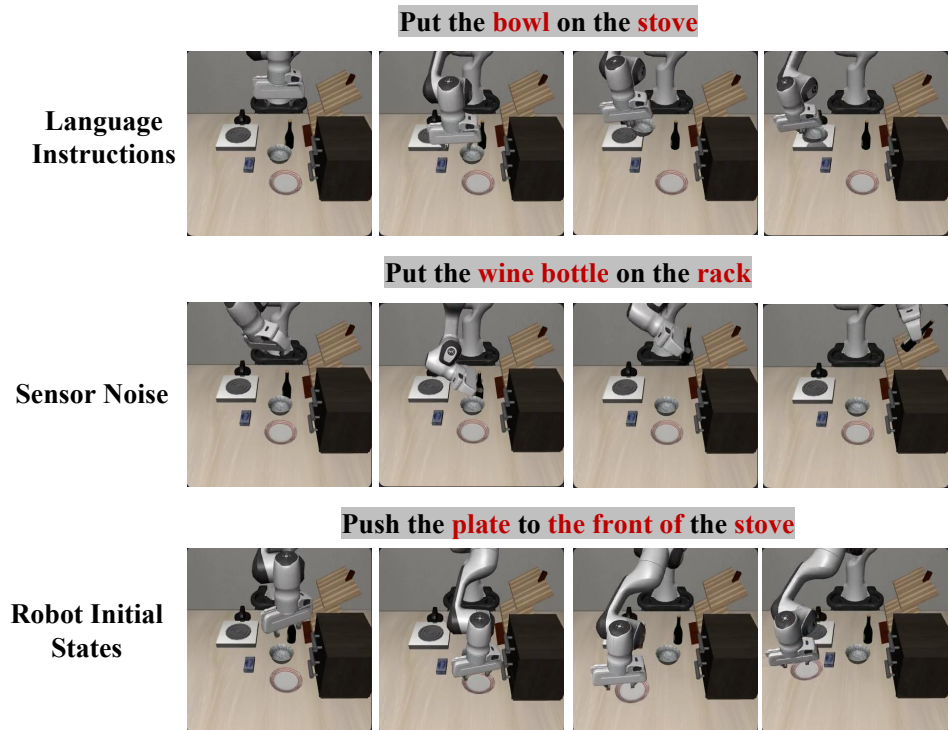


Figure 16: Qualitative results of our CofactVLA model on the **Goal suite of LIBERO-Plus Benchmark**.

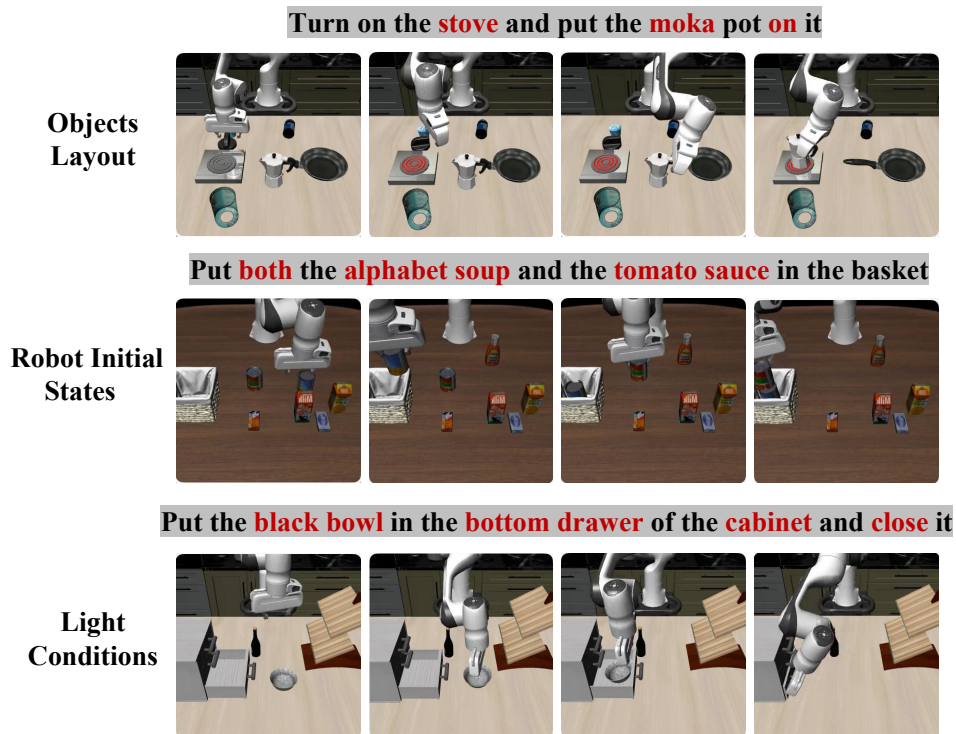


Figure 17: Qualitative results of our CofactVLA model on the **Long suite of LIBERO-Plus Benchmark**.

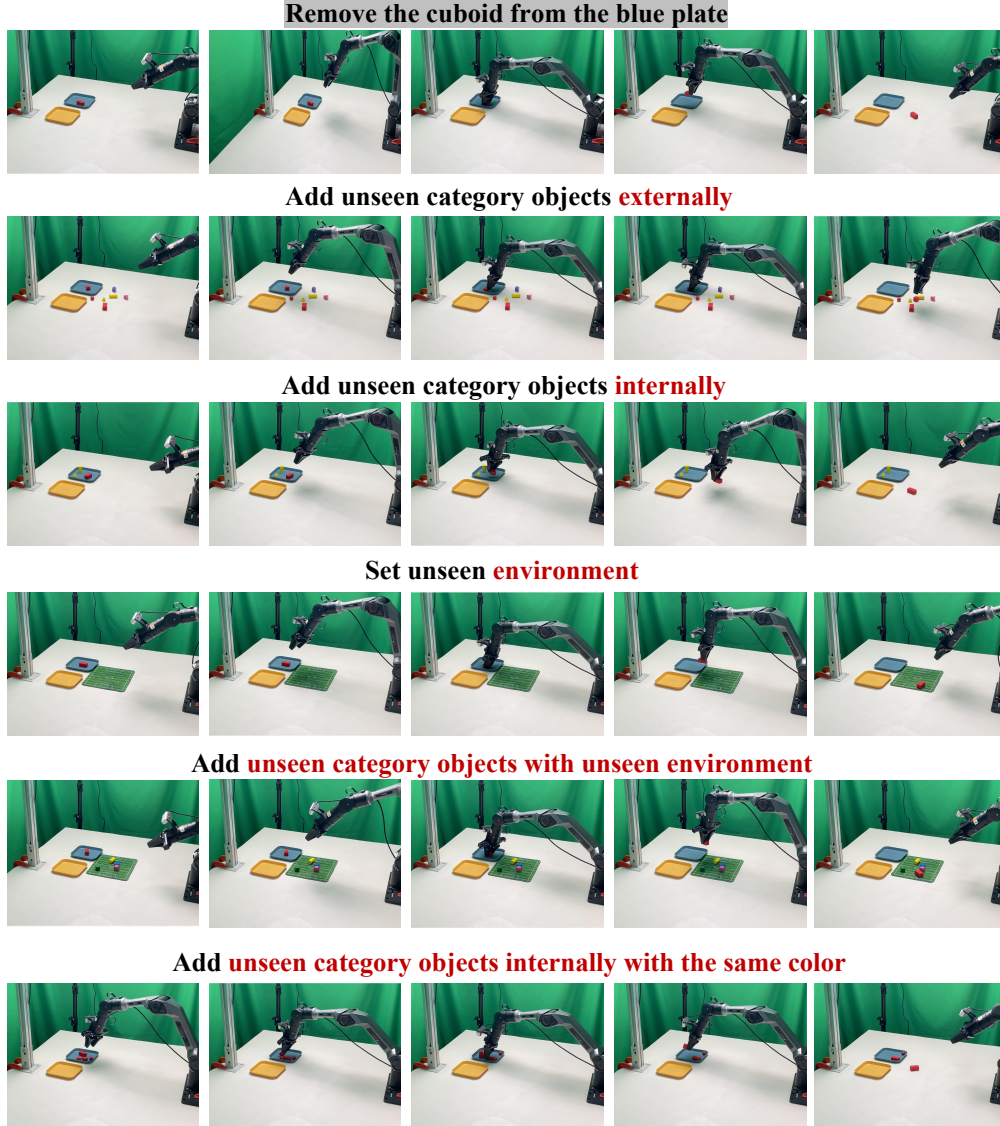


Figure 18: Qualitative results of our CofactVLA model on the **real-world robot experiments**.

physical occlusions—such as when the robot’s end-effector entirely blocks the camera’s field of view (FoV) during precision grasping.

A.7 Limitations and Outlooks

Dependence on Base VLM Priors. First, the overall efficacy of our deconfounding framework is fundamentally bottlenecked by the inherent zero-shot semantic grounding capabilities of the underlying pre-trained Vision-Language Model (VLM). While our causal intervention effectively isolates and neutralizes spurious visual correlations, it cannot artificially synthesize semantic knowledge that the base model fundamentally lacks. Consequently, if the foundational model lacks the latent representations for highly specific visual attributes (e.g., rare object textures or novel geometries), our method could not compensate for this intrinsic knowledge deficit. Therefore, scaling up foundational vision-language pre-training remains a prerequisite to further pushing the boundaries of open-world generalization.

Vulnerability to Severe Visual Occlusions. Second, empirical evaluations in real-world physical deployments highlight a susceptibility to severe visual occlusions. During close-proximity manipulation,

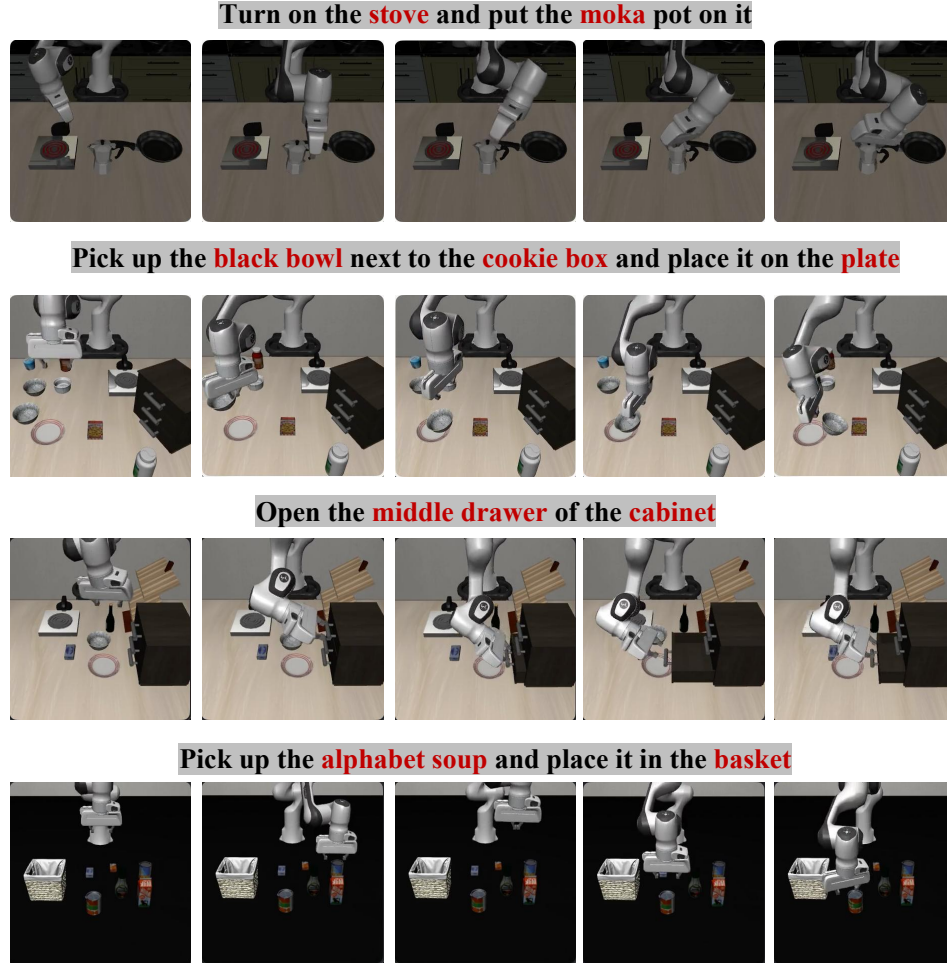


Figure 19: Failed results of our CofactVLA model on the **LIBERO-Plus benchmark**.

the robotic arm or the end-effector may temporarily obstruct the camera’s field of view. This transient loss of critical visual feedback will disrupt the continuous flow-matching generation, occasionally leading to execution failures. Importantly, this stems from a hardware-induced perceptual limitation rather than an inherent algorithmic deficiency in our causal policy. To address this, future research could explore the integration of dynamic multi-view representation learning or real-time 3D scene understanding. Constructing a robust, occlusion-aware perceptual latent space would significantly unlock the full potential of our deconfounded VLA framework in highly cluttered and dynamic physical environments.

A.8 Broader Impacts

The deployment of robust, language-grounded VLA models has significant positive societal potential, particularly in automating household chores, assisting the elderly, and operating in hazardous environments. By mitigating the vision-override phenomenon, CofactVLA ensures that robots strictly adhere to human semantic instructions, thereby enhancing physical safety and reliability. However, potential negative impacts include the acceleration of labor displacement in manual sectors. Furthermore, highly capable open-world manipulation models could theoretically be repurposed for harmful physical tasks if deployed without proper guardrails. We encourage future research to establish rigorous physical safety evaluations and alignment protocols before widespread commercial deployment.

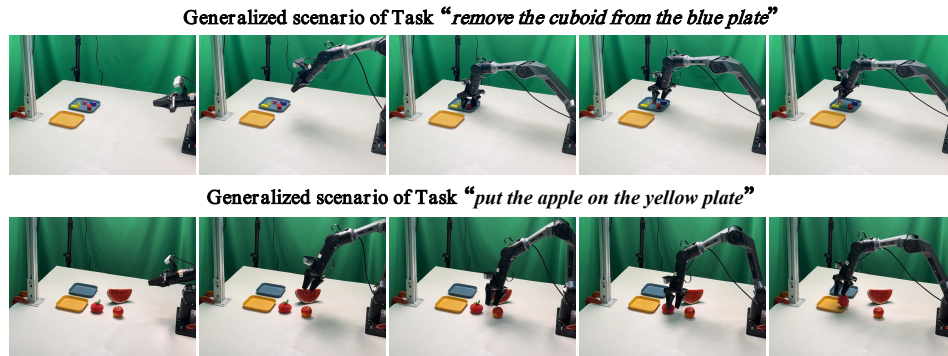


Figure 20: Failed results of our CofactVLA model on the **generalization experiment of real-world deployments**.